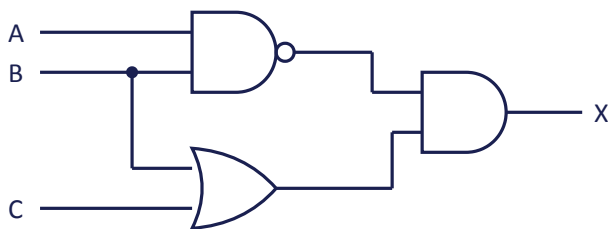


1. Complete the truth table for the logic expression $X = (\text{NOT } A) \text{ AND } (B \text{ OR } C)$. [4]

A	B	C	X
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

2. A logic circuit is shown. Write the logic expression for output X. [3]



X =

3. State the name of the logic gate that: [2]

(a) outputs 0 only when *both* of its inputs are 1

(b) outputs 1 only when *both* of its inputs are 0

4. The truth table of a logic gate is shown. State the name of the gate. [1]

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

Gate:

5. State the number of rows in the truth table of a logic circuit with: [2]

(a) 4 inputs (b) n inputs

6. An OR gate and an XOR gate receive the same two inputs. State the input combination for which the two gates give different outputs. [1]

A = B =

The ladder this week — Karel, not Python. No ice-cream challenge by design: this week's code is **Karel 09 HarvestTheField** and **10 WallFollower**. Gates are the paper half.

Want to dig? Question 3 named two gates, and each of them *on its own* can build every other gate there is — they are the **universal gates**. Prove one: make NOT, then AND, then OR, using nothing but NAND. Then look up how many transistors a NAND costs against an AND, and you will see why real chips are built from the awkward gate rather than the obvious one.

draft / working